Original Article

A consumer-perceived consumer-based brand equity scale

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ABSTRACT **Brand equity is an essential concept in marketing academia and practice. The term came into use during the late 1980s, and the importance of conceptualizing, measuring and managing brand equity has grown rapidly in the eyes of practitioners and academics alike. Despite the importance of the concept, and the need for brand equity measures, the literature lacks an empirically based consumer-perceived brand equity scale. This article develops a brand equity conceptualization and scale determined by dimensions that consumers perceive. This *consumer-perceived consumer-based brand equity scale* is made up of four dimensions: quality, preference, social influence and sustainability. This new robust scale contributes to the theoretical understanding of consumer-based brand equity measurement, and assists brand managers in measuring brand equity and understanding how consumers value brands in order to develop successful brand strategies.**

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INTRODUCTION

Brand equity is an important marketing concept in both academia and practice (Keller and Lehmann, 2006; Christodoulides and de Chernatony, 2010). Consumers develop feelings and associations with brands and become loyal to them due to their 'added value' (Barwise, 1993). Brands have thus become valuable assets to companies, and learning and understanding how to build, measure and manage brand equity is of utmost importance (Kapferer, 1997). The term came into use during the late 1980s, and the importance of conceptualizing,

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measuring and managing brand equity has grown rapidly in the eyes of practitioners and academics alike (for example, Aaker and Keller, 1990; Aaker, 1991, 1996; Keller, 1993; Ailawadi *et al*, 2003; Netemeyer *et al*, 2004; Erdem *et al*, 2006). This has resulted in 'several often-divergent view-points on the dimensions of brand equity, the factors that influence it, the perspectives from which it should be studied, and the ways to measure it' (Ailawadi *et al*, 2003, p. 1). One aspect that academics seem to agree on is that brand equity is a multi-dimensional construct.

Aaker's (1991) and Keller's (1993) conceptualizations of brand equity are the most well-known and commonly adopted. Neither author, though, operationalized a scale for measurement. Moreover, Aaker's (1991) brand equity dimensions are questionable (Christodoulides *et al*, 2015) and 'are accepted largely on the basis of face validity and no attempt is made to argue their relative importance or possible inter-relation' (Shocker, 1993, p. 257). Many academics have thus taken to the task of quantifying brand equity. All have used Aaker (1991) and Keller's (1993) conceptualizations, and have not taken into account what characteristics of a brand are important to consumers – that is, brand equity dimensions. In other words, the literature lacks an empirically based consumer-perceived brand equity scale. Since brands are psychological concepts in consumers' minds (Bedbury and Fenichell, 2002) – that is, brands are what people perceive them to be – the consumer is an active participant or partner in the creation of equity for the brand (Blackston, 2000). Thus, taking into account the dimensions that consumers use to assess the value of a brand is critical to understand, manage, and measure brand equity directly and more effectively.

Furthermore, this approach for assessing the value of a brand seems evermore important in today's marketplace where consumers are actively participating in brand identity creation (Vallaster and von Wallapach, 2013). Past research shows that brand identity is not solely created by companies anymore (Prahalad and Ramaswamy, 2004; Csaba and Bengtsson, 2006; Schau *et al*, 2009; Ind *et al*, 2013). Given that branding has become a collaborative, value co-creation activity (Brodie *et al*, 2006; Merz *et al*, 2009), this article explores the dimensions that consumers use to assess the value of a brand, and develops, validates and proposes a new consumer-perceived consumer-based brand equity scale.

The objective of this article is to develop a brand equity conceptualization and scale determined by dimensions that consumers perceive. Despite the importance of the concept, and the need for brand equity measures, the literature lacks an empirically based consumer-perceived brand equity scale. A multi-step scale development process was adopted in order to develop and validate the *consumer-perceived consumer-based brand equity scale*, which is made up of four dimensions: quality, preference, social influence and sustainability. This new robust scale contributes to the theoretical understanding of consumer-based brand equity measurement, and assists brand managers in measuring brand equity and understanding how consumers value brands in order to develop successful brand strategies.

LITERATURE REVIEW

Brand equity

Although there is no universally accepted definition of brand equity, most authors agree with Farquhar's (1989) definition that brand equity denotes 'the added value with which a given brand endows a product' (p. 24). Strong brand equity leads to opportunities for successful brand extensions, resilience against competitor's promotional efforts, and creation of barriers of entry to competition (Farquhar, 1989) – that is, it provides a competitive advantage. A more formal definition of brand equity is proposed by Ailawadi *et al* (2003, p. 1): 'the outcomes that accrue to a product with its brand name compared with those that would accrue if the same product did not have the brand name'.

Brand equity emanates from three different perspectives. The first is the cognitive psychology perspective, which defines brand equity as the differential consumer response to a brand's marketing mix that results from consumers' associations to a brand (that is, Aaker, 1991; Keller, 1993; Anderson, 2007). The second is the information economics perspective, which defines brand equity as the increased utility that a brand name gives to a product (that is, Wernerfelt, 1988; Erdem and Swait, 1998). The third is the financial markets perspective, which defines brand equity as a financial measure of a firm's market value minus its tangible asset value (that is, Simon and Sullivan, 1993). A fourth category of brand equity has been recently proposed: *employee-based brand equity* is 'the differential effect that brand knowledge has on an employee's response to their work environment' (King and Grace, 2009, p. 130). All these different perspectives are relevant and complementary to each other, as nowadays the value of a brand is created in a participative way by multiple stakeholders (Merz *et al*, 2009; Payne *et al*, 2009; Hatch and Schultz, 2010). Recent research has identified that customer-based brand equity (CBBE) scales do not adequately explain FBBE, and thus call for treating both constructs independently (Nguyen *et al*, 2015). Therefore, this research exclusively focuses on the consumer-based value perspective.

The two most influential conceptualizations of brand equity are Aaker's (1991) and Keller's (1993). Aaker (1991, p.15) defines brand equity as 'a set of brand assets and liabilities linked to a brand, its name and symbol, that add to or subtract from the value provided by a product or service to a firm and/or to that firm's customers'. Aaker proposes four dimensions of brand equity: *brand awareness, brand loyalty, brand associations* and *perceived quality*. Keller (1993) defines CBBE by stating that the power of a brand rests in consumer's minds; on what they have learned, felt, seen and heard about the brand through time. Hence, CBBE is 'the differential effect of brand knowledge on consumer response to the marketing of the brand' (Keller, 1993, p. 2).

Brand equity measurement

Brand equity has been measured according to the three previously discussed perspectives: at the customer level (Aaker, 1991; Keller, 1993; Shocker *et al*, 1994; Lassar *et al*, 1995; Aaker and Joachimsthaler, 2000; Berry, 2000; Chen, 2001; Bendixen *et al*, 2003; Baker *et al*, 2005; Srinivasan *et al*, 2005; Tong and Hawley, 2009; Christodoulides *et al*, 2012), the company or firm level (Farquhar *et al*, 1991; Cobb-Walgren *et al*, 1995; Dyson *et al*, 1996; Kapferer, 1997; Doyle, 2001; Kim *et al*, 2003; Ferjani *et al*, 2009; King and Grace, 2009), and the financial market level (Simon and Sullivan, 1993; Aaker and Jacobson, 1994; Barth *et al*, 1998). Many authors have also developed models that encompass all aspects of brand equity (Srivastava *et al*, 1998; Epstein and Westbrook, 2001; Keller and Lehmann, 2003; Burmann *et al*, 2009).

There are two complementary approaches to measurement: (i) a direct approach, which measures CBBE by assessing the actual impact of brand knowledge on customer response to different marketing elements, and (ii) an indirect approach, which assesses potential sources of CBBE by identifying and tracking customers' brand knowledge structure. Examples of the direct approach include the financial or market-outcome-based measures of brand equity such as revenue premium (Ailawadi *et al*, 2003; Ferjani *et al*, 2009), brand equity as a price premium measure (Holbrook, 1992; Randall *et al*, 1998), and brand equity as a measure of brand extendibility (Randall *et al*, 1998). Examples of the indirect approach include the measurement of brand equity that takes an overall picture of the brand and looks at it through its dimensions. Most authors that have used the indirect approach have developed scales to measure brand equity at different levels and using different contexts (Lassar *et al*, 1995; Yoo and Donthu, 2001; Vazquez *et al*, 2002; de Chernatony *et al*, 2004; Christodoulides *et al*, 2006; Christodoulides *et al*, 2012; Veloutsou *et al*, 2013).

Direct approaches

Srinivasan (1979), Park and Srinivasan, (1994) and Jourdan (2002) use a multi-attribute model to measure consumer-based brand equity. Srinivasan (1979) measures brand equity – 'brand-specific effect' – by comparing actual choice preferences with consumer preferences obtained through conjoint analysis. The difference between overall preference and the preference estimated by the model is quantified into a monetary scale. The limitation of this method is that it provides only segment-level estimates of brand equity that do not shed light on the sources of brand value. Park and Srinivasan (1994) develop a survey-based method for measuring and understanding a brand's equity in a product category. They obtain a brand equity measure at the individual level by dividing brand equity into attribute-based (based on consumers' evaluations of the brand's physical characteristics), and non-attribute-based (based on symbolic associations attached to the brand) components. Their method, however, does not break down the non-attribute based component of brand equity for further insights. Jourdan (2002) improves the reliability and validity of Park and Srinivasan's (1994) measurement. He notes that Park and Srinivasan's calculation of differences of utilities includes an error term. Therefore, Jourdan (2002) uses a repeated measures experimental design to show that an error component is not negligible in this case. Although this method has advantages over its predecessors, the complex experimental design used makes it difficult to put into practice.

Leuthesser *et al* (1995) begin with the assumption that personal evaluation of a given brand on a number of attributes is always biased since consumers are predisposed toward brands they know. This predisposition is manifest through the psychological 'halo effect', which the authors believe is the basis of brand equity. In order to isolate the outcome of the 'halo effect' the authors use two techniques: 'partialling out' and 'double centering'. This method, however, does not indicate the underlying dimensions of consumer-based brand equity, measures equity at the aggregate rather than at the individual level, and is difficult to use by brand managers.

Kamakura and Russell (1993) use a segment-wise logit model to examine consumers' actual purchase behavior. The authors use real purchase data from supermarket checkout counters in their empirical estimation of a model of Brand Value. They go on to identify two major sources of brand equity: Brand Value (which provides a good diagnostic for a brand's competitive position) and Intangible Brand Value (which isolates the utility associated with intangible factors such as brand associations and perceptual distortions). This approach, however, evaluates consumer-based brand equity at the aggregate level, similar to Srinivasan (1979), and can only be used when scanner data is available.

Swait *et al* (1993) build on the information economics paradigm by using the entire value attached to a brand rather than isolating specific parameters. They argue that any measure of brand equity should reflect total utility because the effect of brand equity occurs throughout the components of the utility function. Hence, they propose a measure of consumer-based brand equity called 'Equalization Price' (EP). EP is a measure of the implicit value of a brand in a market where some degree of differentiation exists versus its implicit value in a market characterized by no brand differentiation. Swait *et al* (1993) propose a method that permits the calculation of consumer-based brand equity at the individual level, identifies the sources of brand associations and determines their weighted importance in terms of consumer utility. Nevertheless, the model assumes that all consumers have identical preferences, so unless it is being tested in homogeneous consumer choice markets this method is inappropriate.

Shankar *et al* (2008) develop a model of brand equity based on a customer survey and financial measures. They describe two parts to brand equity, offering value and relative brand importance. The authors propose certain drivers of brand image that can be captured through a consumer survey: brand reputation, brand uniqueness, brand fit, brand associations, brand trust, brand innovation, brand regard and brand fame. This method combines both financial and consumer data, but does not take into consideration that some companies' financial measures may be unavailable at the brand level. In addition, this is an aggregate measure of brand equity because only relative brand importance is measured at the individual level.

Ferjani *et al* (2009) combine a modeling approach and a conjoint methodology to understand the relationships between CBBE and market-based brand equity (MBBE). They argue that consumer preference for a branded product is a result of the incremental profit generated by a branded product versus a private-label product. Without the need to collect consumer brand association data, they present a method that provides dollar-metric values for brand equity while taking into account the effects of brand awareness and market availability as sources of brand equity. Their approach is novel and effective in terms of determining FBBE, but not CBBE.

Indirect approaches

Indirect approaches to the measurement of brand equity take an overall picture of the brand and look at it through its manifest dimensions. For example, Lassar *et al* (1995) propose five CBBE dimensions performance, value, social image, trustworthiness and commitment. They use survey data from consumers collected in two product categories: TV monitors and watches. Their 17-item Likert scale has adequate levels of internal consistency and discriminant validity. However, the scale does not include any behavioral components of brand equity such as loyalty. Also, the authors do not report any tests on the scale's external validity.

Vazquez *et al* (2002), develop a measurement instrument of brand equity that has four basic dimensions: product functional utility, product symbolic utility, brand name functional utility and brand name symbolic utility. Their 22-item scale shows a reasonable degree of reliability and validity for the sports shoes sector and is relatively easy to administer. Kocak *et al* (2007) try to replicate the results of Vazquez *et al* (2002) in Turkey. They use the exact same 22-item scale, which was originally developed and tested in Spain, in the sports shoes sector as well. Kocak *et al* (2007) find that the original scale does not work for the Turkish sample, so they adapt it to 16 items. The authors conclude that the differences between Vazquez *et al* 's (2002) study and their replication are due to cultural differences.

Some category and industry specific measures of consumer-based brand equity have also been developed. de Chernatony *et al* (2004) develop a brand performance measure for financial services brands. Christodoulides *et al* (2006) measure brand equity in an online context and look at the ways that the Internet has allowed consumers to become co-creators of brand value. Davis *et al* (2009) develop a measure of brand equity in the logistics services. Rajasekar and Nalina (2008) and Boo *et al* (2009) develop models of consumer-based brand equity applicable to tourist destinations.

The most robust brand equity scale in the literature to date is Yoo and Donthu's (2001) multi-dimensional brand equity (MBE) scale (Christodoulides and de Chernatony, 2010). Yoo and Donthu (2001) use the four components of CBBE put forth by Aaker (1991). The resulting MBE scale is made up of three dimensions: brand loyalty, brand perceived quality and brand awareness/associations. They also develop a separate scale to measure overall brand equity in order to assess the MBE scale's convergent validity. In the authors' experience using this scale in various studies, the average variance explained was repeatedly low and the loyalty and overall brand equity items loaded under one same factor. There have been many limitations to the use of this and the other above-mentioned scales (Christodoulides *et al*, 2015). Furthermore, a thorough review of the literature reveals that the most widely referred to conceptualizations (that is, Aaker, 1991 and Keller, 1993), and subsequently developed scales, are in fact purely conceptual (see Tables 1 and 2) without confirming or validating their relevance to consumers.

This experience and analysis allowed the authors to confirm that a new measure of consumer-based brand equity is in fact necessary (Yoo and Donthu, 2001; Washburn and Plank, 2002), and, to the authors' surprise, that no effort has been made to fill this gap. Only recently Christodoulides *et al* (2012) suggested an alternative scale to identify the components of consumer-based brand equity. They identify potential brand equity items and dimensions from the perspective of experts in brand management using qualitative research. Although their approach is a step in what we consider is the right direction, it is not a purely consumer-derived brand equity scale. Given the dynamic creation of brand meaning and value in today's marketplace (Brodie *et al*, 2006; Merz *et al*, 2009), this article hence takes a first step to fill this gap, and contributes to the brand equity literature, by developing a fully consumer-based and market oriented brand equity scale. With data from representative samples of consumers in the United States it, first, understands the consumers' perspective of brand equity through qualitative research and, second, identifies scale items that measure what consumers actually perceive to be the relevant dimensions of brand equity (that is, brand characteristics that they perceive make a brand more valuable to them).

SCALE DEVELOPMENT

In order to develop and validate a scale measuring consumer-perceived brand equity, a multi-step process (DeVellis, 1991) was followed. This is a modified version of the frameworks proposed by Parasuraman *et al* (1988) and Churchill (1979).

Identification of the dimensions of consumer-perceived brand equity

Qualitative data was collected from 423 undergraduate marketing students at a University in the Southwest of the United States (Gummesson, 2005). Students were

Table 1: Direct approach brand equity scales

Direct approach							
#	*Author*	*Year*	*Use*	*Description*	*Base conceptualization*	*Dimensions*	*Definition of brand equity*
1	Srinivasan	1979	Measurement	Use the multi-attribute model to measure consumer-based brand equity.	N.A.	N.A.	'Brand-specific effect' is the component of a brand's overall prefernce that is not explained by the multi-attribute model
2	Kamakura and Russell	1993	Scale	Looked at perceived quality and brand intangible value of CBBE	N.A.	Perceived quality, brand intangible value	The implied utility or value assigned to a brand by consumer
3	Swait *et al*	1993	Scale	EP as a measure of brand equity	N.A.	N.A.	Propose a measure of consumer-based brand equity called EP which is the monetary expression of the utility a consumer attributes to a bundle consisting of a brand name, product attributes and price
4	Park and Srinivasan	1994	Measurement	Achieved measurement of brand equity at the individual level	N.A.	Attribute-based brand equity, non-attribute-based brand equity	The difference between an individual consumer's overall brand preference and his or her multi-attributed preference based in objectively measured attribute levels
5	Leuthesser *et al*	1995	Scale	The halo effect measure of brand equity	Keller (1993)	N.A.	From Keller: brand equity represents the value to a consumer of a product, above that which would result for otherwise identical product without the brand's name
6	Jourdan	2002	Measurement	Amendment and improvement of the Park and Srinivasan measurement model using experimental design	Park and Srinivasan (1994)	Attribute-based brand equity, non-attribute-based brand equity	The difference between the subjective preference and the objective preference *vis-à-vis* the product
7	Ailawadi *et al*	2003	Scale	Propose and validate revenue premium as an outcome measure of brand equity	Keller (1993)	Revenue premium is the difference in revenue (net price×volume) between a branded good and a corresponding private label	From various authors: the marketing effects or outcomes that accrue to a product with its brand name compared with those that would accrue if the same product did not have the brand name
8	Shankar *et al*	2008	Scale	A multi-category brand equity model and its application	N.A.	Offering value (net present value or financial worth of an offering carrying a brand name) and RBI (relative brand importance derived from consumer brand choice and determined by brand image and other marketing-mix elements)	From Shocker and Weitz (1988): the net present value of the incremental cash flows attributable to a brand name and to the firm owning that brand relative to an identical product with no brand name or band-building efforts

Table 2: Indirect approach brand equity scales

					Indirect approach	
#	*Author*	*Year*	*Description*	*Base conceptualization*	*Dimensions*	*Definition of Brand Equity*
1	Lassar *et al*	1995	Measuring CBBE	Keller (1993)	Performance, social image, value, trust worthiness and attachment	Brand equity stems from the greater confidence that consumers place in a brand than they do in its competitors. This confidence translates into consumers' loyalty and their willingness to pay a premium price for the brand
2	Yoo and Donthu	2001	MBE scale	Aaker (1991, 1996) and Keller (1993)	Brand loyalty, perceived quality and brand awareness/associations	Consumers' different response between a focal brand and an unbranded product when both have the same level of marketing stimuli and product attributes
3	Vazquez *et al*	2002	Developing and validating a measurement instrument for consumer-based brand equity	Kamakura and Russell (1993) and Cobb-Walgren *et al* (1995)	Product utility (product functional utility, product symbolic utility) and brand name utility (brand name functional utility, brand name symbolic utility)	The overall utility that the consumer associates to the use and consumption of the brand; including associations expressing both functional and symbolic utilities
4	Washburn and Plank	2002	Modifications of the Yoo and Donthu (2001) scale	Aaker (1991, 1996) and Keller (1993)	Brand loyalty, perceived quality and brand awareness/associations	Consumers' different response between a focal brand and an unbranded product when both have the same level of marketing stimuli and product attributes
5	de Chernatony *et al*	2004	Developed a brand performance measure for financial services brands	Depth interviews	Brand loyalty, satisfaction, reputation	Marketing Science Institute (MSI) definition of Brand Equity
6	Netemeyer *et al*	2004	Developing and validating measures of facets of CBBE	Aaker (1991, 1996) and Keller (1993)	Perceived quality, perceived value for the cost, uniqueness, and the willingness to pay a price premium for a brand	From Keller: CBBE occurs when the consumer is familiar with the brand and holds some favorable, strong and unique associations in memory
7	Pappu *et al*	2005	An improvement to the measurement of consumer-based brand equity	Aaker (1991, 1996) and Keller (1993)	Brand awareness, brand associations, perceived quality and brand loyalty	From Farquhar: value endowed by the brand to the product
8	Christodoulides *et al*	2006	Conceptualizing and measuring the equity of online brands	Qualitative research	Emotional connection, online experience, responsive service nature, trust and fulfillment	Online Retail/Service (ORS) Brand equity is defined as a relational type of intangible asset that is co-created through the interaction between consumers and the e-tail brand
9	Kocak *et al*	2007	Replication of the consumer-based equity scale developed by Vazquez *et al* (2002)	Vasquez *et al* (2002)	Product utility and brand name utility (revised scale from 22 to 16 items)	From Vazquez *et al*: overall utility that consumer associates to the use and consumption of the brand, including associations expressing both functional and symbolic utilities

10	Guizani *et al*	2009	Working paper; development of a scale for consumer brand equity using French consumers (non-students)	Aaker (1991, 1996) and Keller (1993)	Brand loyalty, perceived brand quality, brand knowledge and social value	From Farquhar: value added by the brand name to the product
11	Lehmann *et al*	2008	Suggest a parsimonious set of brand measures that can be used to measure brand performance	Keller and Lehmann (2003)	Comprehension, comparative advantage, interpersonal relations, history , preference and attachment	Brand performance can be thought of in terms of four stages: awareness, image and associations, preference, and attachment
12	Rajasekar and Nalina	2008	A new measure of CBBE in India	Aaker (1996)	Performance, social image, value, trust worthiness and attachment	From Farquhar: value endowed by the brand to the product
13	Buil *et al*	2008	New brand equity scale including personality dimensions (items from many different authors)	Aaker (1991)	Brand awareness, perceived quality, brand loyalty and brand associations (perceived value, brand personality and organizational associations)	From Aaker: a set of brand assets and liabilities linked to a brand, its name and symbol that add to or subtract from the value provided by a product or service to a firm and/or to that firm's customers
14	Zeugner-Roth *et al*	2008	Country-brand equity scale	Yoo and Donthu (2001)	Country brand loyalty, perceived country brand quality, and country brand awareness/associations	From Farquhar: define country brand equity as the value-added brought forth by the association of a product or brand with a given country name, as perceived by the individual consumer
15	Davis *et al*	2009	Measuring brand equity for logistics services	Keller (1993)	Brand awareness, brand image, overall brand equity	From Aaker: a set of brand assets and liabilities linked to a brand, its name and symbol that add to or subtract from the value provided by a product or service to a firm and/or to that firm's customers
16	Boo *et al*	2009	Developed a destination brand scale (used Las Vegas and Atlantic City as destinations)	Aaker (1991, 1996) and Keller (1993)	Destination brand awareness, destination brand image, destination brand quality, destination brand value and destination brand loyalty	Overall utility that customers place in a brand compared with its competitors
17	Atilgan *et al*	2009	Emergence of brand trust as a new dimension instead of brand awareness	Aaker (1991, 1996) and Keller (1993)	Perceived quality, brand loyalty, brand associations and brand trust	From Yoo and Donthu (2001): difference in consumer choice between the focal branded product and an unbranded product given the same level of product features and Keller (1993): brand with equity provide an ownable, trustworthy, relevant, distinctive promise to consumers
18	Christodoulides *et al*	2012	Suggest an alternative scale to identify components of CBBE from the perspective of experts in brand management	Aaker (1991)	Awareness, heritage, uniqueness, reliability and willingness to sacrifice	From Farquhar: brand equity is a key intangible asset that arises from past brand building activities and encompasses the added value endowed by the brand to the product

asked to respond to the following question: 'In 5–10 words please state what first comes to your mind when you think of a brand that is important to you'. The question was asked in a broad and general manner, and without any reference to a specific brand, as a first exploratory step to identify what makes brands valuable to consumers. Given that students did not provide more than 5 words each, frequency count was used to identify the most commonly reported words. The word that was provided by students the most was perceived value, 296 times. Following, the words were grouped together with similar terms based on common underlying themes to identify a set of preliminary consumer-perceived dimensions (Gummesson, 2005). Four preliminary dimensions were identified: *perceived quality, perceived value, brand preference* and *sustainability*. Perceived quality describes how consumers perceive a brand in terms of its quality, consistency, performance, reliability, functionality and being well made. Perceived value describes how consumers perceive a brand's value in terms of price/cost, affordability, value, ease of use and availability. Brand preference is a description of consumers' perception of their loyalty to a brand in terms of how trusting, honest, likeable, comfortable and dependable they believe the brand is. Finally, sustainability describes how consumers are more aware of and loyal to brands that are environmentally safe/responsible, sustainable and healthy.

Since the first sample used student responses to come up with dimensions, in order to validate these dimensions, additional qualitative data were collected from a nationwide representative consumer sample of the US population using an omnibus survey. One thousand consumers were asked to respond to the same broad question. Consumers provided on average of 2 words each. Frequency count was again used to identify the most commonly reported words, which were then grouped together based on common underlying themes (Gummesson, 2005). The most commonly provided answer was quality, 356 times (see Table 3 for a list of words and their frequency counts). The result of this process confirmed the preliminary consumer-perceived brand equity dimensions (see Table 4).

Table 3: Qualitative results

Words	*Frequency*
Quality	356
Reliable	121
Price	106
Value	84
Taste	64
Trust	62
Consistent	43
Dependable	36
Affordable	35
Availability	28
Works well	24
Safety	22
Cost	21
Durable	18
Loyalty	18
Healthy	17
Reputation	15
Ease of use	14
Performance	14
Likability	9
Comfortable	8
Honesty	8
Long-lasting	8
Meets needs	8
Environmentally responsible	5

Perceived quality and brand preference are dimensions that have similar items to Aaker and Keller's conceptualizations. However, perceived value incorporates other aspects that consumers find important in their evaluation of a brand – that is, price, value, ease-of-use and availability. In addition, sustainability brings to light a perspective that consumers have toward their brand choices today. Consumers are aware of the level of social responsibility they expect from a brand and perceive this in terms of how safe, sustainable, healthy, reputable and environmentally responsible the brand is.

Table 4: Perceived dimensions from qualitative study

Perceived quality	*Perceived value*	*Brand preference*	*Sustainability*
• Quality • Reliable • Consistent • Works well • Durable • Long-lasting • Performance	• Price • Cost • Affordable • Value • Ease of use • Available	• Loyalty • Trust • Honesty • Likeable • Comfortable • Dependable	• Safety • Sustainable • Healthy • Reputation • Environmentally responsible

Item generation and scale purification

Scale items were identified to measure each of the brand characteristics that made up the four preliminary dimensions (see Table 4). For perceived quality, two scale items, used by Yoo and Donthu (2001), were adapted and four scale items were generated to measure the additional attributes consumers delineated in the qualitative research. For perceived value, Sweeney and Soutar's (2001) consumer-perceived value scale was adapted and modified to fit the context of this research. For brand preference, the three loyalty scale items used by Yoo and Donthu (2001) were adapted, as well as one scale item on commitment from Berry (2000). Four scale items were generated to measure the additional attributes identified in the qualitative research. For sustainability, Brown and Dacin (1997) social responsibility scale was adapted, and five new scale items were generated to measure the additional attributes identified in the qualitative research. The list of original items in the survey can be found in Table 5.

On the basis of suggestions by Tull and Hawkins (1994), Bliss *et al* (1987), Aaker and Day (1990) and DeVellis (1991), the scale items were given to two academic experts on scale development and brand equity for their perusal and comments. This, as noted by Parasuraman *et al* (1988), was designed to ensure the elimination of ambiguous statements before the first stage of data collection. The result added one scale item, suggested by an expert, to the preference dimension 'I am committed to buying (X)' from Berry (2000).

The initial data collection was composed of undergraduate marketing students. The self-administered online survey was carried out using a non-probability convenience sample of 90 students attending a large University in the Southwest of the United States. The product category purposely chosen for this survey was electronics, specifically smart phones. This product category is widely understood and valued by both students and non-students. Furthermore, consumers have well-established perceptions regarding this product category. In addition, three widely-known, heavily advertised brands, with clearly established brand identities were used: iPhone, BlackBerry and HTC. These specific brands were tested and chosen due to their high brand recognition for the population being surveyed at the time.

According to Churchill (1979), Parasuraman *et al* (1988), Tull and Hawkins (1994), Dabholkar *et al* (1996) and Fabrigar *et al* (1999), exploration of the underlying structure of the data was carried out through exploratory factor analysis (EFA) (Hair *et al*, 2009). Principal component analysis (Nunnally, 1978) using varimax rotation was used. An eigenvalue greater than 1 and a cumulative percentage of variance explained greater than 60 per cent were used as criteria in determining the number of factors. On the basis of these criteria, 6 factors were extracted, which collectively accounted for an

Table 5: Original items in survey

Original items in survey	*Author*
Brand preference	
I consider myself to be loyal to (X)	Yoo and Donthu (2001)
(X) would be my first choice	Yoo and Donthu (2001)
I will not buy other brands if (X) is available at the store	Yoo and Donthu (2001)
I feel that (X) is trustworthy	Qualitative research
(X) is an honest brand	Qualitative research
I feel comfortable when I buy (X)	Qualitative research
(X) is a dependable brand	Qualitative research
I am committed to buying (X)	Berry (2000)
Perceived quality	
The quality of (X) is extremely high	Yoo and Donthu (2001)
The functionality of (X) is very high	Yoo and Donthu (2001)
The reliability of (X) is very high	Qualitative research
(X) is consistent in the quality it offers	Qualitative research
The durability of (X) is very high	Qualitative research
The performance of (X) is very high	Qualitative research
Perceived value	
(X) has consistent quality	Sweeney and Soutar (2001)
(X) is well made	Sweeney and Soutar (2001)
(X) has an acceptable standard of quality	Sweeney and Soutar (2001)
(X) has poor workmanship	Sweeney and Soutar (2001)
(X) would *not* last a long time	Sweeney and Soutar (2001)
(X) performs consistently	Sweeney and Soutar (2001)
(X) is a brand I would enjoy	Sweeney and Soutar (2001)
(X) would make me want to use it	Sweeney and Soutar (2001)
(X) is a brand that I would feel relaxed about using	Sweeney and Soutar (2001)
(X) would make me feel good	Sweeney and Soutar (2001)
(X) would be economical	Sweeney and Soutar (2001)
(X) is reasonably priced	Sweeney and Soutar (2001)
(X) offers value for money	Sweeney and Soutar (2001)
(X) is a good product for the price	Sweeney and Soutar (2001)
(X) would help me to feel acceptable	Sweeney and Soutar (2001)
(X) would improve the way I am perceived	Sweeney and Soutar (2001)
(X) would make a good impression on other people	Sweeney and Soutar (2001)
(X) would give its owner social approval	Sweeney and Soutar (2001)
Sustainability	
(X) is an environmentally safe brand	Qualitative research
(X) is a sustainable brand	Qualitative research
(X) is a healthy brand	Qualitative research
(X) is a brand with a good reputation	Qualitative research
(X) is an environmentally responsible brand	Qualitative research
(X) is a socially responsible brand	Brown and Dacin (1997)
(X) is more beneficial to society's welfare than other brands	Brown and Dacin (1997)
(X) does not contribute something to society	Brown and Dacin (1997)

average variance explained of 85 per cent between the 3 brands. At this point, only initial testing of the internal reliability of the extracted factors was performed in the form of Cronbach's coefficient alpha (α). The cut-off value adopted was 0.70 and all factors were above the cut-off point (see Churchill, 1979; Parasuraman *et al*, 1988; Spector, 1992). Following the examination of the EFA results and analysis of the reliability and conceptual coherency of the indicated factors, the six identified factors

Table 6: Composition of the samples

Brand	*Sample*	*# of respondents*	*Gender*		*Age*		*Education*	
			Male	*Female*	*< = 30*	*> 30*	*< = High school*	*> High school*
iPhone	Consumers	138	75	63	34	104	34	104
BlackBerry	Consumers	138	59	79	34	104	42	96
HTC	Consumers	127	61	66	29	98	27	100
iPhone	Students	163	75	88	150	13	10	153
BlackBerry	Students	162	71	91	153	9	13	149
HTC	Students	152	60	92	145	7	8	144
		880	46%	54%	62%	38%	15%	85%

were: *quality, preference, sustainability, social influence, durability* and *perceived value*. Given that the pre-test was the first time this survey was used, the emergence of multiple dimensions is not surprising given its exploratory nature. Furthermore, the sample size was relatively small (90 responses), so a larger sample size should yield a clearer picture of the dimensions. It is important to note, though, that all four dimensions that originated in the qualitative research were re-validated in this pre-test (quality, preference, sustainability and perceived value).

Data collection

In order to test and validate the initial factors identified, a second survey using two different samples; a student sample and a consumer sample, was used. Data were collected simultaneously from the two samples (477 students and 403 consumers). The student sample was made up of marketing undergraduate students, 57 per cent of which were female, 94 per cent were less than 30 years old, and 94 per cent had higher than a high school degree. The consumer group was a nation-wide representative sample and data was collected using an omnibus survey. In the consumer sample, 52 per cent were female, 24 per cent were less than 30 years old and 74 per cent had higher than a high school degree. Care was taken to make sure that each of the three brands (iPhone, BlackBerry and HTC) received equal responses in both population groups (see Table 6).

Assessment of the reliability and validity of the scale

The proposed dimensions were assessed in terms of reliability and validity (Spector, 1992). The analysis for all three brands (see Tables 7, 8 and 9) yielded a 21-item factor structure made up of four factors: quality, preference, sustainability and social influence. The scale was reduced from six to four dimensions due to the fact that the items in the previously labeled 'perceived value' and 'durability' dimensions did not meet the .70 Cronbach's α cut-off. Therefore, the resultant four dimensions were labeled: *quality, preference, sustainability* and *social influence*.

The student sample and the consumer sample were analyzed separately. For the student sample, the descriptive statistics can be found in Table 10. The reliability coefficients for the resulting factors are high for all three brands (ranging from 0.70 to 0.97). In addition, the inter-item correlations for the 21 items are acceptable with each item above the cut-off point of 0.30. The descriptive statistics for the consumer sample can be found in Table 11. Confirmatory factor analysis (CFA) was employed for both samples to test the measurement structure and examine how well the data set fits the hypothesized

Table 7: Exploratory factor analysis – iphone (student sample)

	Quality	*Social influence*	*Preference*	*Sustainability*
The quality of iPhone is extremely high	0.863	—	—	—
iPhone has consistent quality	0.852	—	—	—
iPhone has an acceptable standard of quality	0.847	—	—	—
The performance of iPhone is very high	0.813	—	—	—
iPhone performs consistently	0.812	—	—	—
The reliability of iPhone is very high	0.805	—	—	—
The functionality of iPhone is very high	0.798	—	—	—
iPhone is consistent in the quality it offers	0.791	—	—	—
iPhone is well made	0.775	—	—	—
iPhone would make a good impression on other people	—	0.882	—	—
iPhone improves the way I am perceived by others	—	0.879	—	—
iPhone would give its owner social approval	—	0.879	—	—
iPhone helps me feel accepted	—	0.836	—	—
I am committed to buying iPhone	—	—	0.843	—
I will not buy other brands if iPhone is available at the store	—	—	0.836	—
I consider myself to be loyal to iPhone	—	—	0.828	—
iPhone would be my first choice	—	—	0.796	—
iPhone is an environmentally safe brand	—	—	—	0.834
iPhone is an environmentally responsible brand	—	—	—	0.812
iPhone is a healthy brand	—	—	—	0.812
iPhone is a sustainable brand	—	—	—	0.737
Total Variance Explained = 83.197%	56.986	11.38	8.318	6.512
Cronbach's α	0.98	0.964	0.948	0.923

Table 8: Exploratory factor analysis – BlackBerry (student sample)

	Quality	*Preference*	*Sustainability*	*Social Influence*
The reliability of BlackBerry is very high	0.897	—	—	—
BlackBerry is consistent in the quality it offers	0.894	—	—	—
The performance of BlackBerry is very high	0.877	—	—	—
The quality of BlackBerry is extremely high	0.864	—	—	—
The functionality of BlackBerry is very high	0.840	—	—	—
BlackBerry has consistent quality	0.835	—	—	—
BlackBerry performs consistently	0.828	—	—	—
BlackBerry has an acceptable standard of quality	0.819	—	—	—
BlackBerry is well made	0.808	—	—	—
BlackBerry would be my first choice	—	0.907	—	—
I consider myself to be loyal to BlackBerry	—	0.891	—	—
I will not buy other brands if BlackBerry is available at the store	—	0.889	—	—
I am committed to buying BlackBerry	—	0.850	—	—
BlackBerry is an environmentally safe brand	—	—	0.850	—
BlackBerry is an environmentally responsible brand	—	—	0.819	—
BlackBerry is a sustainable brand	—	—	0.779	—
BlackBerry is a healthy brand	—	—	0.767	—
BlackBerry improves the way I am perceived by others	—	—	—	0.867
BlackBerry would make a good impression on other people	—	—	—	0.838
BlackBerry would give its owner social approval	—	—	—	0.821
BlackBerry helps me feel accepted	—	—	—	0.729
Total Variance Explained = 81.794%	51.6	12.877	10.042	7.274
Cronbach's α	0.975	0.931	0.951	0.954

Table 9: Exploratory factor analysis – HTC (student sample)

	Quality	*Social influence*	*Preference*	*Sustainability*
The quality of HTC is extremely high	0.838	—	—	—
The functionality of HTC is very high	0.834	—	—	—
The performance of HTC is very high	0.829	—	—	—
HTC has an acceptable standard of quality	0.809	—	—	—
The reliability of HTC is very high	0.800	—	—	—
HTC is well made	0.800	—	—	—
HTC has consistent quality	0.768	—	—	—
HTC is consistent in the quality it offers	0.730	—	—	—
HTC performs consistently	0.679	—	—	—
HTC improves the way I am perceived by others	—	0.891	—	—
HTC helps me feel accepted	—	0.843	—	—
HTC would give its owner social approval	—	0.834	—	—
HTC would make a good impression on other people	—	0.812	—	—
I will not buy other brands if HTC is available at the store	—	—	0.880	—
HTC would be my first choice	—	—	0.873	—
I consider myself to be loyal to HTC	—	—	0.861	—
I am committed to buying HTC	—	—	0.797	—
HTC is an environmentally safe brand	—	—	—	0.874
HTC is an environmentally responsible brand	—	—	—	0.827
HTC is a sustainable brand	—	—	—	0.750
HTC is a healthy brand	—	—	—	0.667
Total variance explained = 82.072%	56.818	10.183	8.436	6.635
Cronbach's α	0.979	0.963	0.95	0.965

Table 10: Descriptive statistics (student sample)

	BlackBerry – Students		*HTC – Students*		*iPhone – Students*	
	Mean	*Standard deviation*	*Mean*	*Standard deviation*	*Mean*	*Standard deviation*
The reliability of (Brand X) is very high	4.17	1.48	4.39	1.447	5.42	1.427
(Brand X) is consistent in the quality it offers	4.2	1.483	4.54	1.451	5.55	1.419
The performance of (Brand X) is very high	4.09	1.495	4.56	1.473	5.7	1.384
The quality of (Brand X) is extremely high	4.22	1.499	4.51	1.483	5.63	1.379
The functionality of (Brand X) is very high	4.19	1.541	4.62	1.509	5.77	1.283
(Brand X) has consistent quality	4.19	1.526	4.6	1.348	5.42	1.503
(Brand X) performs consistently	4.15	1.443	4.42	1.379	5.37	1.401
(Brand X) has an acceptable standard of quality	4.4	1.463	4.73	1.395	5.63	1.41
(Brand X) is well made	4.34	1.467	4.75	1.42	5.53	1.48
(Brand X) would be my first choice	1.98	1.506	2.85	1.76	4.73	2.261
I consider myself to be loyal to (Brand X)	1.98	1.594	2.84	1.791	4.17	2.364
I will not buy other brands if (Brand X) is available at the store	1.83	1.336	2.55	1.508	4.07	2.267
I am committed to buying (Brand X)	2.14	1.544	3.06	1.484	4.33	2.241
(Brand X) is an environmentally safe brand	3.96	1.065	4.08	1.077	4.37	1.324
(Brand X) is an environmentally responsible brand	4.03	1.106	4.16	1.101	4.52	1.326
(Brand X) is a sustainable brand	4.07	1.298	4.39	1.213	4.96	1.488
(Brand X) is a healthy brand	4.02	1.182	4.24	1.276	4.87	1.424
(Brand X) improves the way I am perceived by others	3.05	1.637	3.62	1.474	3.79	2.044
(Brand X) would make a good impression on other people	3.35	1.68	3.89	1.555	4.23	1.936
(Brand X) would give its owner social approval	3.36	1.74	3.78	1.569	4.18	1.95
(Brand X) helps me feel accepted	3.01	1.609	3.61	1.532	3.9	1.946

Table 11: Descriptive statistics (consumer sample)

	BlackBerry – Consumers		*HTC – Consumers*		*iPhone – Consumers*	
	Mean	*Standard deviation*	*Mean*	*Standard deviation*	*Mean*	*Standard deviation*
The reliability of (Brand X) is very high	4.47	1.466	4.66	1.356	4.91	1.632
(Brand X) is consistent in the quality it offers	4.54	1.461	4.65	1.381	4.94	1.593
The performance of (Brand X) is very high	4.48	1.525	4.66	1.391	4.93	1.601
The quality of (Brand X) is extremely high	4.55	1.44	4.65	1.416	4.88	1.679
The functionality of (Brand X) is very high	4.57	1.528	4.66	1.339	4.99	1.634
(Brand X) has consistent quality	4.47	1.446	4.72	1.377	4.93	1.541
(Brand X) performs consistently	4.46	1.357	4.67	1.3	4.85	1.533
(Brand X) has an acceptable standard of quality	4.57	1.445	4.79	1.358	5	1.509
(Brand X) is well made	4.55	1.357	4.74	1.402	5.04	1.549
(Brand X) would be my first choice	2.94	1.799	3.83	1.58	3.7	2.156
I consider myself to be loyal to (Brand X)	2.51	1.764	3.73	1.618	3.15	2.085
I will not buy other brands if (Brand X) is available at the store	2.72	1.678	3.72	1.579	3.43	2.086
I am committed to buying (Brand X)	2.9	1.613	3.76	1.588	3.62	2.108
(Brand X) is an environmentally safe brand	4.08	1.313	4.34	1.17	4.28	1.273
(Brand X) is an environmentally responsible brand	4.07	1.242	4.35	1.152	4.37	1.34
(Brand X) is a sustainable brand	4.14	1.332	4.43	1.2	4.7	1.374
(Brand X) is a healthy brand	4.02	1.359	4.37	1.14	4.45	1.42
(Brand X) improves the way I am perceived by others	3.36	1.584	3.94	1.405	3.86	1.837
(Brand X) would make a good impression on other people	3.72	1.552	4.16	1.304	4.1	1.765
(Brand X) would give its owner social approval	3.67	1.577	4.05	1.337	3.99	1.866
(Brand X) helps me feel accepted	3.52	1.59	3.96	1.334	3.8	1.86

Table 12: Reliability and factor structure (LISREL) diagnostics

Brand	*Sample*	*Cronbach's α*	χ^2	*RMSEA*	*GFI*	*AGFI*	*NFI*	*NNFI*	*CFI*	*IFI*	*RFI*	*RMR*
iPhone	Consumer	0.949	624.14	0.108	0.72	0.66	0.95	0.96	0.97	0.97	0.94	0.059
BlackBerry	Consumer	0.948	608.66	0.106	0.73	0.66	0.95	0.97	0.97	0.97	0.94	0.067
HTC	Consumer	0.951	721.59	0.128	0.67	0.59	0.92	0.94	0.95	0.95	0.91	0.091
iPhone	Student	0.952	435.65	0.072	0.82	0.77	0.96	0.98	0.98	0.98	0.95	0.041
BlackBerry	Student	0.939	557.49	0.091	0.78	0.72	0.94	0.96	0.97	0.97	0.93	0.071
HTC	Student	0.950	505.72	0.086	0.78	0.73	0.95	0.97	0.97	0.97	0.94	0.053

measurement structure (see Dabholkar *et al*, 1996). This was undertaken via the Lisrel structural equation modeling approach. In line with Hair *et al* (2009), model goodness-of-fit was evaluated through (i) absolute-fit measures, (ii) incremental-fit measures and (iii) parsimonious-fit measures. Individual indices examined for the first fit measures were Likelihood-ratio chi-square (*P*-value), Goodness-of-fit Index (GFI), and Root of Mean Square Error of Approximation (RMSEA) value; for the second fit, the Adjusted Goodness-of-fit Index (AGFI), Normed Fit Index (NFI) and Non-Normed Fit Index (NNFI) were examined. As for the third fit, the Comparative Fit Index (CFI), Incremental Fit Index (IFI), Relative Fit Index (RFI) and Root Mean Square Residual (RMR) were examined (see Hair *et al*, 2009).

The results of the analysis of the model goodness-of-fit for both samples are shown in Table 12. Although the chi-square (χ^2) values were statistically significant at the

0.05 level (a common occurrence, given the well-known sensitivity of this statistic (Bagozzi and Yi, 1988), other structure diagnostics (that is, NFI, NNFI, CFI, IFI and RFI) were generally supportive and appeared to fall within the acceptable ranges (Hair *et al*, 2009). Reliability was assessed by using Cronbach's α, which ranged from 0.939–0.952, and therefore, all samples have a reliability greater than 0.70 (Nunnally, 1978). Consequently, the results indicate appreciable goodness-of-fit; and hence, it is judged that the structure is well supported for iPhone, BlackBerry and HTC for both samples.

Assessing content, convergent, predictive, nomological and discriminant validity

Content validity

The measures developed for the consumer-based brand equity factors were derived from an exhaustive systematic approach that commenced with examination of the relevant literature, a two-step qualitative data collection process, followed by the generation of statements. The measures underwent detailed evaluations by both academicians and experts in the industry thus providing the confirmation for content validity. Furthermore, during the refinement of the questionnaires, a pre-test was conducted among a group of undergraduate students. The results of these approaches indicated that the content of the factors was well represented by the measurement items. It must be mentioned, however, that although the judgment of content validity is subjective, the procedures used in this research are consistent with good practice in the literature (see Churchill, 1979; DeVellis, 1991).

Convergent validity

Convergent validity was examined through CFA where the average extracted variance can be seen as a set of common underlying dimensions of the research construct – that is, factors explaining consumer-perceived consumer-based brand equity.

Predictive validity

The considerable degree of inter-factor correlations and evidence of internal reliability (see Cronbach α's for all scales in Table 12) attest to the stability of this scale in terms of predictive validity. Thus, overall, the four factors identified predict consumer-perceived consumer-based brand equity.

Nomological and discriminant validity

Given that this scale uses a combination of validated scale items from the extant literature and the qualitative research, the resulting consistency with extant scales in brand equity indicates nomological validity. The average variances extracted and squared interconstruct correlations were satisfactory (Fornell and Larcker, 1981). All phi's squared were consistently lower than the reliability estimates. This procedure satisfies the requirement of discriminant validity (Hair *et al*, 2009).

Identification of the new consumer-perceived brand equity scale

The proposed 21-item scale has acceptable fit, with acceptable discriminant and convergent validity, and internal consistency reliability and parsimony (see Table 13). Therefore, the consumer-perceived brand equity scale is made up of four dimensions: *quality, preference, social influence* and *sustainability*.

DISCUSSION

The scale to measure consumers'-perceived consumer-based brand equity developed in this study is made up of four dimensions

Table 13: The new consumer-perceived brand equity scale

Quality	The reliability of (Brand X) is very high
	(Brand X) is consistent in the quality it offers
	The performance of (Brand X) is very high
	The quality of (Brand X) is extremely high
	The functionality of (Brand X) is very high
	(Brand X) has consistent quality
	(Brand X) performs consistently
	(Brand X) has an acceptable standard of quality
	(Brand X) is well made
Preference	(Brand X) would be my first choice
	I consider myself to be loyal to (Brand X)
	I will not buy other brands if (Brand X) is available at the store
	I am committed to buying (Brand X)
Sustainability	(Brand X) is an environmentally safe brand
	(Brand X) is an environmentally responsible brand
	(Brand X) is a sustainable brand
	(Brand X) is a healthy brand
Social influence	(Brand X) improves the way I am perceived by others
	(Brand X) would make a good impression on other people
	(Brand X) would give its owner social approval
	(Brand X) helps me feel accepted

(quality, preference, social influence and sustainability) each supported by four to nine items. The scale dimensions discussed in the ensuing sections are labeled according to different characteristics consumers value in a brand, through which the brand acquires its equity.

Quality

Quality is 'the consumer's judgment about a product's overall excellence or superiority' (Zeithaml, 1988, p. 3). Quality – brand quality – describes how consumers perceive a brand in terms of its consistency, acceptable standards, performance, reliability, functionality and being well made. Quality as a dimension was included in the previous conceptualizations of Aaker (1991) and Keller (1993). However, the quality dimension in this new scale contains consumer derived items that have never before been included together in any brand equity scale.

Preference

A preferred brand commands loyalty. Preference describes how a brand is a consumer's first choice, how the consumer is loyal to a specific brand, how a consumer will not buy other brands if that specific brand is available at the store, and how a consumer is committed to buying that brand. Loyalty is a factor that was previously included in the Aaker (1991) and Keller (1993) conceptualizations. However, the proposed preference dimension is based on consumer derived items identified in the qualitative research phase of this study, and measured by combining scale items from Yoo and Donthu (2001) and Berry (2000), allowing for conceptualizing this new factor.

Social influence

Social influence is a dimension that has never been included in any previous brand equity scale. Consumers perceive that a brand improves the way they are perceived by others, makes a good impression on other people, gives its owner social approval and helps them feel accepted. The four consumer derived items that make up the dimension of social influence were measured with scale items from Sweeney and Soutar's (2001) consumer-perceived value scale. Although these items do in fact signify value to the consumers, in this case, this dimension describes ways in which consumers gain value by using (purchasing) a brand in order to achieve social approval (Thompson and Arsel, 2004) and build their *persona* (Kapferer, 1997).

Sustainability

The sustainability dimension is fully made up of consumer perceived items identified in the qualitative research of this study. Hence, the scale items used for measuring them were newly generated for the purpose of this research: '(X) is an environmentally safe brand', '(X) is an environmentally responsible brand', '(X) is a sustainable brand' and '(X) is a healthy brand'. Nowadays, consumers are more aware of brands and companies that are environmentally safe, environmentally responsible, sustainable and healthy. A consumer is more loyal to a brand and believes it is of higher quality if it is sustainable (Simmons and Becker-Olsen, 2006; Nan and Heo, 2007; Lai *et al*, 2010). Sustainability is becoming part of the brand identity of many products and brands that consumers purchase on a daily basis. Companies such as Toms Shoes, Ben and Jerry's and Warby Parker are good examples of brands that have integrated sustainability and social responsibility into their mission. Sustainability is thus a current and important dimension that has not been included in any brand equity scale before.

THEORETICAL AND MANAGERIAL CONTRIBUTIONS

This research identifies consumer-perceived dimensions that make up consumer-based brand equity. The proposed scale is simple to use and will be valued by academics and practitioners alike. This work not only follows Churchill's (1979) scale development method, but also enhances it by merging qualitative with quantitative data from the consumer. The CFA supports the goodness-of-fit of the data set and the hypothesized structure. The study contributes to the literature and adds to the broad discussion on current conceptualizations and measures of brand equity. Previous studies have provided results of consumer-based brand equity based on Aaker's (1991) and Keller's (1993) conceptualizations and using students or convenience samples to test their scales, leading to a lack in methodological rigor (see Tables 1 and 2). This is the first study that develops a consumer-perceived conceptualization and scale that is tested and validated with both student and consumer samples.

This study benefits brand research in several ways. The scale can be used to re-assess and measure the equity of brands from a consumer derived perspective, and to analyze how brand equity's potential antecedents impact how consumers value a brand. However, the impact of each of the scale's dimensions on consequent variables needs to be identified. Further research could focus on studying the nomological relationships between consumer-perceived consumer-based brand equity and new identified variables. The scale can also be used to aid in studies of brand value and brand extensions. For example, it serves to measure the effectiveness of a co-branding effort by measuring the equity of each brand separately and then measuring the brand equity as a co-brand. We can also use this new measure in testing the perceived sustainability of different brands in different product categories. This could in turn lead to new positioning strategies for many brands.

Also, there may be a potential causal order among the dimensions of brand equity. According to Levidge and Steiner (1961), the hierarchy of effects model suggests that perceived quality precedes brand loyalty – brand preference in the new scale. Hence, if there is a strong order among the dimensions, then in order to manage resources more efficiently, managers should consider strategies that place focus on each of the four dimensions at different times, or as part of a phased approach.

Another important use for this scale is that it can possibly be used to rank brands based on their perceived consumer value. In other words, a brand equity index can be determined to rank brands according to the four dimensions (that is, a brand might be high or low on each of the dimensions). Ranking brands could allow for new segmentation strategies. Given that the products from the category used, smart phones, are expensive, and have short useful lives – from a branding perspective – , the proposed scale seems suitable for measuring the brand equity of other goods and consumer personal accessories, ranging from clothing to consumer beverages and furniture to automobiles, whose value has been widely established in consumer markets. Further research could assess the scale's effectiveness for other product categories.

In order to enhance the measurement of brand equity from a consumer perspective, the findings of this study encourage managers to re-assess their brand equity with respect to quality, preference, social influence and sustainability. Most firms, due to a lack of availability of a good scale, have been measuring brand equity using one-dimensional measures. The proposed scale is valid, reliable and also parsimonious. Its ease of use will allow managers to track the equity of their brands on a regular basis.

LIMITATIONS AND FUTURE RESEARCH DIRECTIONS

Although the study described to this point makes a contribution to the marketing literature and serves as a springboard for future research, some considerations are necessary. First, although a consumer sample was used in the study, a student sample was also employed as a comparison sample. This student sample encompasses all of the requisite caveats regarding representativeness and generalizability. Also, respondents from this student sample were geographically concentrated in the Southwestern United States, and they were younger than the population at large. However, students of this age group do represent a segment of consumers who purchase smart phones (that is, iPhone, BlackBerry and HTC).

There was also a limitation in the sample size as the consumer sample and student sample were limited in the number of respondents (student sample = 477, consumer sample = 403). These limitations pose generalizability questions without further replication and validation. Although our attempt is consistent with the literature and directions from academic experts, we acknowledge the lack of (true experimental) control over the identified factors. One exception to this is the use of a consumer sample in addition to the student sample to test and validate the scale. However, a larger sample would be more appropriate for future studies to re-validate the scale.

Another limitation of this study is that only one product category was used. As a result, participants saw one of three brands, again limiting generalizability. This research should encourage researchers to revise and revalidate the scale. It would be desirable to replicate this study using additional product categories and brands. However, limiting the study allowed the research to limit, and better control for, extraneous and confounding variables. Finally, developing a scale that is based on qualitative findings, although methodologically sound, implies that the scale may be dynamic over time. As consumer values evolve, what they find to be valuable in a brand may also evolve. Therefore, future research will help revise and re-validate the scale over time and categories.

In addition to replication studies with other product categories and brands, this study points to a number of interesting directions for future research. Some research directions include: testing and validating the scale using a larger nation-wide sample and comparing that to nation-wide samples in various other countries (cross-cultural studies), testing and validating the scale using other product categories and brands including services and industrial goods, testing the dimensions to see if they may be extended into sub-dimensions, using the identified factors to determine if any of them can help explain how consumers respond to brand/product-crises, testing a second order CFA model to assess the causal order among the dimensions to identify brands' strengths, and testing the relationship between financial brand equity and this consumer-perceived brand equity.

In summary, the four dimensions of the consumer-perceived brand equity scale are: quality, preference, social influence and sustainability. These dimensions measure brand equity effectively with a total explained variance of at least 80 per cent and good reliability and validity using both a student sample and a nation-wide consumer sample. The scale that is developed and validated in this study not only contributes to the marketing discipline's body of knowledge, but it also provides a springboard for future research into brand equity. The value of a consumer-perceived brand equity scale suggests a number of new directions for study and elaboration in what is certain to be a compelling stream of research with vast implications for both theory and practice.

REFERENCES

Aaker, D.A. (1991) *Managing Brand Equity*. New York: Free Press.

Aaker, D.A. (1996) *Building Strong Brands*. New York: Free Press.

Aaker, D.A. and Day, G.S. (1990) *Marketing Research*, 4th edn. Toronto, ON: John Wiley & Sons.

Aaker, D.A. and Jacobson, R. (1994) The financial information content of perceived quality. *Journal of Marketing Research* 31(2): 191–201.

Aaker, D.A. and Joachimsthaler, E. (2000) *Brand Leadership*. New York: Free Press.

Aaker, D.A. and Keller, K.L. (1990) Consumer evaluations of brand extensions. *Journal of Marketing* 54(1): 27–41.

Ailawadi, K., Lehmann, D.R. and Neslin, S.A. (2003) Revenue premium as an outcome measure of brand equity. *Journal of Marketing* 67(4): 1–17.

Anderson, J. (2007) Brand equity: The perpetuity perspective, in *Winter 2007* Proceedings of the American Marketing Association, Vol. 18, San Diego, CA, p. 142.

Atilgan, E., Akinci, S., Aksoy, S. and Kaynak, E. (2009) Customer-based brand equity for global brands: A multinational approach. *Journal of Euromarketing* 18(2): 115–132.

Bagozzi, R.P. and Yi, Y. (1988) On the evaluation of structural equation models. *Journal of Academy of Marketing Science* 16(1): 74–94.

Baker, C., Nancarrow, C. and Tinson, J. (2005) The mind versus market share guide to brand equity. *International Journal of Marketing Research* 47(5): 525–542.

Barth, M.E., Clement, M.B., Foster, G. and Kasznik, R. (1998) Brand values and capital market valuation. *Revenue of Accounting Studies* 3(1–2): 41–68.

Barwise, P. (1993) Introduction to the special issue on brand equity. *International Journal of Research in Marketing* 10(1): 3–8.

Bedbury, S. and Fenichell, S. (2002) *A New Brand World: 8 Principles for Achieving Brand Leadership in the 21st Century*. New York: Viking.

Bendixen, M., Bukasa, K.A. and Abratt, R. (2003) Brand equity in the business-to-business market. *Industrial Marketing Management* 33(5): 371–380.

Berry, L. (2000) Cultivating service brand equity. *Journal of the Academy of Marketing Science* 28(1): 128–137.

Blackston, M. (2000) Building brand equity by managing the brand's relationships. *Journal of Advertising Research* 32(3): 101–105.

Bliss, J., Monk, M. and Ogborn, J. (1987) *Qualitative Data Analysis for Educational Research*. London: Croom Helm.

Boo, S., Busser, J. and Baloglu, S. (2009) A model of customer-based brand equity and its application to multiple destinations. *Tourism Management* 30(2): 219–231.

Brodie, R.J., Glynn, M.S. and Little, V. (2006) The service brand and the service-dominant logic: Missing fundamental premise or the need for stronger theory? *Marketing Theory* 6(3): 363–379.

Brown, T.J. and Dacin, P.A. (1997) The company and the product: Corporate associations and consumer product responses. *Journal of Marketing* 61(1): 68–84.

Buil, I., de Chernatony, L. and Martinez, E. (2008) A cross-national validation of the consumer-based brand equity scale. *Journal of Product and Brand Management* 17(6): 384–392.

Burmann, C., Jost-Benz, M. and Riley, N. (2009) Towards an identity-based brand equity model. *Journal of Business Research* 62(3): 390–397.

Chen, A.C.H. (2001) Using free association to examine the relationship between characteristics of brand associations and brand equity. *Journal of Product and Brand Management* 10(7): 439–451.

Christodoulides, G., Cadogan, J.G. and Veloutsou, C. (2015) Consumer-based brand equity measurement: Lessons learned from an international study. *International Marketing Review* 32(3/4): XX.

Christodoulides, G., Cadogan, J.G., Veloutsou, C. and de Chernatony, L. (2012) Revisiting brand equity: Evidence from three European countries, in 41st European Marketing Academy Conference (EMAC), 22–25 May, Lisbon, Portugal.

Christodoulides, G. and de Chernatony, L. (2010) Consumer-based brand equity conceptualization and measurement. *International Journal of Market Research* 52(1): 43–66.

Christodoulides, G., de Chernatony, L., Furrer, O. and Abimbola, T. (2006) Conceptualising and measuring the equity of online brands. *Journal of Marketing Management* 22(7/8): 799–825.

Churchill Jr G.A. (1979) A paradigm for developing better measures of marketing constructs. *Journal of Marketing Research* 16(1): 64–73.

Cobb-Walgren, C.J., Ruble, C. and Donthu, N. (1995) Brand equity, brand preference, and purchase intent. *Journal of Advertising* 24(3): 25–40.
Csaba, F. and Bengtsson, A. (2006) Rethinking identity in brand management. In: J.E. Schroeder and M. Salzer-Mörling (eds.) *Brand Culture*. London: Routledge, pp. 118–135.
Dabholkar, P.A., Thorpe, D.I. and Rentz, J.O. (1996) A measure of service quality for retail stores: Scale development and validation. *Journal of the Academy of Marketing Science* 24(1): 3–16.
Davis, D.F., Golicic, S.L. and Marquardt, A. (2009) Measuring brand equity for logistics services. *The International Journal of Logistics Management* 20(2): 201–212.
de Chernatony, L., Harris, F.J. and Christodoulides, G. (2004) Developing a brand performance measure for financial services brands. *Services Industries Journal* 24(2): 15–33.
DeVellis, R.F. (1991) *Scale Development: Theory and Applications*. Newbury Park, CA: Sage.
Doyle, P. (2001) Shareholder-value-based brand strategies. *Journal of Brand Management* 9(1): 20–30.
Dyson, P., Farr, A. and Hollis, N.S. (1996) Understanding, measuring, and using brand equity. *Journal of Advertising Research* 36(6): 9–21.
Epstein, M.J. and Westbrook, R.A. (2001) Linking actions to profits in strategic decision making. *MIT Sloan Management Review* 42(3): 39–49.
Erdem, T., Swait, J. and Valenzuela, A. (2006) Brands as signals: A cross country validation study. *Journal of Marketing* 70(1): 34–49.
Erdem, T. and Swait, J. (1998) Brand equity as a signaling phenomenon. *Journal of Consumer Psychology* 7(2): 131–157.
Fabrigar, L.R., Wegener, D.T., MacCallum, R.C. and Strahan, E.J. (1999) Evaluating the use of exploratory factor analysis in psychological research. *Psychological Methods* 4(3): 272–299.
Farquhar, P.H. (1989) Managing brand equity. *Marketing Research* 1(3): 24–33.
Farquhar, P.H., Han, J.Y. and Iriji, Y. (1991) *Recognizing and Measuring Brand Assets*. Report Cambridge, MA: Marketing Science Institute, pp. 91–119.
Ferjani, M., Jedidi, K. and Jagpal, S. (2009) A conjoint approach for consumer- and firm-level brand valuation. *Journal of Marketing Research* 46(6): 846–862.
Fornell, C. and Larcker, D.F. (1981) Evaluating structural equation models with unobservable variables and measurement error. *Journal of Marketing Research* 18(1): 39–50.
Guizani, H., Trigueiro, H. and Valette-Florence, P. (2009) Development of a scale for French consumer brand equity, working paper in the Latin American Advances in Consumer Research, The University of Waikato, New Zealand, Vol. 2, pp. 198–200.
Gummesson, E. (2005) Qualitative research in marketing: Road-map for a wilderness of complexity and unpredictability. *European Journal of Marketing* 39(3/4): 309–327.
Hair Jr J.F., Black, W.C., Babin, B.J. and Anderson, R.E. (2009) *Multivariate Data Analysis*, 7th edn. Upper Saddle River, NJ: Pearson Prentice Hall.
Hatch, M.J. and Schultz, M. (2010) Towards a theory of brand co-creation with implications for brand governance. *Journal of Brand Management* 17(8): 590–604.
Holbrook, M.B. (1992) Product quality, attributes, and brand name as determinants of price: The case of consumer electronics. *Marketing Letters* 3(1): 71–83.
Ind, N., Iglesias, O. and Schultz, M. (2013) Building brands together: Emergence and outcomes of co-creation. *California Management Review* 55(3): 5–26.
Jourdan, P. (2002) Measuring brand equity: Proposal for conceptual and methodological improvements. *Advances in Consumer Research* 29(1): 290–298.
Kamakura, W.A. and Russell, G.J. (1993) Measuring brand value with scanner data. *International Journal of Research in Marketing* 10(1): 9–22.
Kapferer, J. (1997) *Strategic Brand Management*. London: Kogan Page.
Keller, K.L. (1993) Conceptualizing, measuring, and managing customer-based brand equity. *Journal of Marketing* 57(1): 1–22.
Keller, K.L. and Lehmann, D.R. (2003) How do brands create value? *Marketing Management* 12(3): 26–31.
Keller, K.L. and Lehmann, D.R. (2006) Brands and branding: Research findings and future priority. *Marketing Science* 25(6): 740–759.
Kim, H., Kim, W.G. and An, J.A. (2003) The effect of consumer-based brand equity on firms' financial performance. *Journal of Consumer Marketing* 20(4): 335–351.
King, C. and Grace, D. (2009) Employee based brand equity: A third perspective. *Services Marketing Quarterly* 30(2): 122–147.
Kocak, A., Abimbola, T. and Ozer, A. (2007) Consumer brand equity in a cross-cultural replication: An evaluation of a scale. *Journal of Marketing Management* 23(1/2): 157–173.
Lai, C., Chiu, C., Yang, C. and Pai, D. (2010) The effects of corporate social responsibility on brand performance: The mediating effect of industrial brand equity and corporate reputation. *Journal of Business Ethics* 95(3): 457–469.
Lassar, W., Mittal, B. and Arun, S. (1995) Measuring customer-based brand equity. *Journal of Consumer Marketing* 12(4): 11–19.
Leuthesser, L., Kohli, C.S. and Harich, K.R. (1995) Brand equity: The halo effect measure. *European Journal of Marketing* 29(4): 57–66.
Levidge, R.J. and Steiner, G.A. (1961) A model of predictive measurement of advertising effectiveness. *Journal of Marketing* 25(6): 59–62.
Merz, M.A., He, Y. and Vargo, S.L. (2009) The evolving brand logic: A service-dominant logic perspective. *Journal of the Academy of Marketing Science* 37(3): 328–344.
Nan, X. and Heo, K. (2007) Consumer responses to corporate social responsibility (CSR) initiatives. *Journal of Advertising* 36(2): 63–74.

Netemeyer, R. *et al.* (2004) Developing and validating measures of facets of customer-based brand equity. *Journal of Business Research* 57(2): 209–224.

Nguyen, T.D., Dadzie, C., Davari, A. and Guzmán, F. (2015) Intelectual capital through the eyes of the consumer. *Journal of Product and Brand Management* 24(6): 554–566.

Nunnally, J.C. (1978) *Psychometric Theory*. New York: McGraw-Hill.

Pappu, R., Quester, P.G. and Cooksey, R.W. (2005) Consumer-based brand equity: Improving the measurement – empirical evidence. *Journal of Product and Brand Management* 14(3): 143–154.

Parasuraman, A., Zeithaml, V.A. and Berry, L.L. (1988) SERVQUAL: A multi-item scale for measuring consumer perceptions of service quality. *Journal of Retailing* 64(1): 12–40.

Park, C.S. and Srinivasan, V. (1994) A survey-based method for measuring and understanding brand equity and its extendibility. *Journal of Marketing Research* 31(5): 271–288.

Payne, A., Storbacka, K., Frow, P. and Knox, S. (2009) Co-creating brands: Diagnosing and designing the relationship experience. *Journal of Business Research* 62(3): 379–389.

Prahalad, C.K. and Ramaswamy, V. (2004) Co-creation experiences: The next practice in value creation. *Journal of Interactive Marketing* 18(3): 5–14.

Rajasekar, N. and Nalina, K.G. (2008) Measuring customer-based brand equity in durable goods industry. *Journal of Marketing and Communication* 4(1): 48–58.

Randall, T., Ulrich, K. and Reibstein, D. (1998) Brand equity and vertical product line extent. *Marketing Science* 17(4): 356–379.

Schau, H.J., Muñiz Jr A.M. and Arnould, E.J. (2009) How brand community practices create value. *Journal of Marketing* 73(5): 30–51.

Shankar, V., Azar, P. and Fuller, M. (2008) BRAN*EQT: A multicategory brand equity model and its application at Allstate. *Marketing Science* 27(4): 567–584.

Shocker, A.D. (1993) New books in review: Managing brand equity. *Journal of Marketing Research* 30(2): 256–258.

Shocker, A.D., Srivastava, R.K. and Rueckert, R.W. (1994) Challenges and opportunities facing brand management: An introduction to the special issue. *Journal of Marketing Research* 31(2): 149–158.

Shocker, A.D. and Weitz, B. (1988) A perspective on brand equity principles and issues. In: L. Leuthesser (ed.) *Report number 88-104*. Cambridge, MA: Marketing Science Institute, pp. 2–4.

Simmons, C.J. and Becker-Olsen, K. (2006) Achieving marketing objectives through social sponsorships. *Journal of Marketing* 70(4): 154–169.

Simon, C.J. and Sullivan, M.W. (1993) The measurements and determinants of brand equity: A financial approach. *Marketing Science* 12(1): 28–52.

Spector, P.E. (1992) *Summated Rating Scale Construction*. SAGE University Papers London: SAGE Publications.

Srinivasan, V. (1979) Network models for estimating brand-specific effects in multi attribute marketing models. *Management Science* 25(1): 11–21.

Srinivasan, V., Park, C.S. and Chang, D.R. (2005) An approach to the measurement analysis, and prediction of brand equity and its sources. *Management Science* 51(9): 1433–1448.

Srivastava, R.K., Shervani, T.A. and Fahey, L. (1998) Market-based assets and shareholder value: A framework for analysis. *Journal of Marketing* 62(1): 2–18.

Swait, J., Erdem, T., Louviere, J. and Dubelaar, C. (1993) The equalization price: A measure of consumer-perceived brand equity. *International Journal of Research in Marketing* 10(1): 23–45.

Sweeney, J.C. and Soutar, G.N. (2001) Consumer perceived value: The development of a multiple item scale. *Journal of Retailing* 77(2): 203–220.

Thompson, C.J. and Arsel, Z. (2004) The Starbucks brandscape and consumer (anticorporate) experiences of glocalization. *Journal of Consumer Research* 31(3): 631–642.

Tong, X. and Hawley, J.M. (2009) Measuring customer-based brand equity: Empirical evidence from the sportswear market in China. *Journal of Product and Brand Management* 18(4): 262–271.

Tull, D.S. and Hawkins, D.I. (1994) *Marketing Research: Measurement and Method*. Englewood Cliffs, NJ: Prentice Hall.

Vallaster, C. and von Wallapach, S. (2013) An online discursive inquiry into the social dynamics of multi-stakeholder brand meaning co-creation. *Journal of Business Research* 66(9): 1505–1515.

Vazquez, R., Del Rio, A.B. and Iglesias, V. (2002) Consumer-based brand equity: Development and validation of a measurement instrument. *Journal of Marketing Management* 18(1/2): 27–48.

Veloutsou, C., Christodoulides, G. and de Chernatony, L. (2013) A taxonomy of measures for consumer-based brand equity: Drawing on the views of managers in Europe. *Journal of Product and Brand Management* 22(3): 238–248.

Washburn, J.H. and Plank, R.E. (2002) Measuring brand equity: An evaluation of a consumer-based brand equity scale. *Journal of Marketing Theory and Practice* 10(1): 46–61.

Wernerfelt, B. (1988) Umbrella branding as a signal of new product quality: An example of signaling by posting a bond. *RAND Journal of Economics* 19(3): 458–466.

Yoo, B. and Donthu, N. (2001) Developing and validating a multidimensional consumer based brand equity scale. *Journal of Business Research* 52(1): 1–14.

Zeithaml, V.A. (1988) Consumer perceptions of price, quality and value: A means end model and synthesis of evidence. *Journal of Marketing* 52(3): 2–22.

Zeugner-Roth, K.P., Diamantopoulos, A. and Montesinos, M.A. (2008) Home country image, country brand equity and consumers' product preferences: An empirical study. *Management International Review* 48(5): 577–602.